Nguyen Ba Long

Contact

+84 396 978 406

longnguyen132005@gmail.com

Thu Duc, Ho Chi Minh

https://github.com/NBasLongz

Portfolio.NguyenBaLong

www.linkedin.com/in/nbl2005

Education

- University of Information Technology (UIT) - VNUHCM
- Bachelor of Science in Computer Science (2023 – Feb. 2027, Expected)
- GPA: 8.24/10.0
- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Mining

Skills & More

- Professional Skills: Analytical Thinking, Problem Solving, Self-Directed Research, Team Collaboration.
- Languages: Vietnamese (Native), English (Professional Reading, TOEIC S&W 250).
- Interests: Open-source AI agents, emerging tech, sports, and music.

Summary

Third-year Computer Science student with a solid foundational knowledge and hands-on experience through various course projects in Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Generative AI (NLP/LLMs). Familiar with data preprocessing, building neural network architectures, training models, and designing intelligent RAG/Agent workflows. Seeking an AI/ML Intern opportunity to apply academic knowledge, learn from practical challenges, and continuously improve technical skills.

Experience

AIC Video Retrieval System — *Github*

SigLIP2, Milvus, Elasticsearch, PaddleOCR, Whisper, Flask — *4 - 6/2026*

- Built a multi-modal video retrieval pipeline for the AI Challenge by integrating SigLIP2 visual features, PaddleOCR text, Whisper transcripts, and VLM captions to enable high-accuracy natural language search.
- Developed a hybrid search engine by combining Milvus dense vector search with Elasticsearch sparse matching to improve retrieval rankings using weighted reciprocal rank fusion (RRF).
- Created video preprocessing and reranking workflows using OpenCV and BAAI/bge-reranker-v2-m3 to ensure precise frame-to-millisecond mapping and optimize evaluation payloads.

RAG-based Lecture Video Q&A System — *Github*

LangGraph, Qwen2-VL, Whisper, FastAPI, RAGAS — *3 - 5/2026*

- Collaborated in a 3-member team to build a multi-modal lecture video Q&A system by developing a directed state graph RAG pipeline with LangGraph and asynchronous FastAPI event loops to ensure efficient query routing.
- Synchronized multimodal lecture contents by combining slide text from Qwen2-VL with audio transcriptions from Whisper-large-v3 to accurately map learning materials to exact video timestamps.
- Optimized retrieval performance by implementing a hybrid search engine (BM25 and dense search) with a Cross-Encoder rerank layer, using the RAGAS framework to evaluate answer quality.

Self-Refined RL Reward Designer — *Github*

Python, Flask, Q-Learning, LLM API — *3 - 6/2026*

- Reproduced and adapted a research paper on using Large Language Models (LLMs) to automatically generate and iteratively self-refine reward functions for Reinforcement Learning.
- Remade the complex original environment into a lightweight 2D environment (Gridworld) to enable faster experimentation and CPU-friendly training.
- Built an automated pipeline using Flask API to orchestrate the LLM workflow: sending prompts, extracting Python reward code, executing it in the 2D environment, and sending feedback logs back to the LLM.
- Implemented Prompt Engineering techniques to ensure LLM outputs valid Python code, handling syntax errors and runtime exceptions to maintain a stable self-refinement loop.

Technical Skills

- Languages: Python, C/C++, SQL.
- AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers.
- NLP, LLMs & RAG: LangChain, LangGraph, BM25, Vector Search, Hybrid Retrieval, Cross-Encoder Reranking.
- Backend & Databases: FastAPI, Flask, PostgreSQL, Milvus, Elasticsearch, Docker, Git.